ANALYSIS II — SHEET 2 (for Week 3 tutorials)
Limits and Continuity on General Domains. Limit Points.

1.(i) Consider the function $\mathbf{F}\colon \mathbb{R} \to \mathbb{R}^2$ where $\mathbf{F}(x) = (f(x), g(x))$. Show that $\mathbf{F}$ is continuous if and only if $f, g\colon \mathbb{R} \to \mathbb{R}$ are continuous.

(ii) Show directly that the function $m\colon \mathbb{R}^2 \to \mathbb{R}$ given by $m(x, y) = xy$ is continuous.

(iii) Say that $f, g\colon \mathbb{R} \to \mathbb{R}$ are continuous. Deduce (without AOL) that their product fg is continuous.

2. The function $F\colon \mathbb{R}^2 \to \mathbb{R}$ is defined by $F(0,0) = 0$ and

$$F(x,y) = \frac{xy}{x^2+y^2}, \qquad (x,y) \neq (0,0).$$

Show that F is not continuous at $(0,0)$. Show, however, for each $a \in \mathbb{R}$, the functions $f_a(x) = F(x,a)$ and $g_a(y) = F(a,y)$ are continuous on all of $\mathbb{R}$.

3. (i) Give an example of an infinite set $S \subseteq \mathbb{R}$ with no limit points.
(ii) Give an example of an infinite set $S \subseteq \mathbb{R}$ with countably infinite limit points.
(iii) Give an example of an infinite set $S \subseteq \mathbb{R}$ with uncountably many limit points.
(iv) Show that if $S \subseteq \mathbb{R}$ is infinite and bounded then S has a limit point.
(v) Show that if $S \subseteq \mathbb{R}$ is uncountable then S has a limit point.

4. Set $C_0 = [0,1]$ and inductively define

$$C_{n+1} = \frac{1}{3}C_n \cup \left(\frac{1}{3}C_n + \frac{2}{3}\right).$$

The *Cantor set* $C \subseteq [0,1]$ is then defined as

$$C = \bigcap_{n=0}^{\infty} C_n.$$

(i) Show that C_n is a union of 2^n closed intervals of length 3^{-n}.
(ii) Show that every point of C is a limit point of C.
(iii) Show that C is uncountable. [Hint: consider the base 3 expansions of elements in C.]

5. Given $\varnothing \neq A \subseteq \mathbb{R}$, define $f_A\colon \mathbb{R} \to \mathbb{R}$ by

$$f_A(x) = \inf\{|x-y| : y \in A\}.$$

(i) Sketch a graph of f_A when $A = (0,1) \cup (2,3]$.
(ii) Show that f_A is a well-defined, continuous function on $\mathbb{R}$.
(iii) Show that $f_A(x) = 0$ if and only if $x \in A$ or x is a limit point of A.
(iv) Given $\varnothing \neq B \subseteq \mathbb{R}$, show that

$$f_{A\cup B}(x) = \min\{f_A(x), f_B(x)\}.$$

6. Let $f\colon S \to \mathbb{R}$ and p be a limit point of S. Suppose, for each sequence p_n in S converging to p, that the sequence $f(p_n)$ converges. Show that the limits of the sequences $f(p_n)$ are all equal. Deduce that if $f(x)$ does not converge as $x \to p$ then there is a sequence p_n in S such that $p_n \to p$ and $f(p_n)$ does not converge.

Extension Exercises (Optional)

7. (i) Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be continuous. Show that $f^{-1}(0)$ is a closed subset of $\mathbb{R}^2$.
(ii) Determine which of the following subsets $S_i \subseteq \mathbb{R}^2$ are closed subsets.

$$\begin{aligned} S_1 &= \left\{(x,y) \in \mathbb{R}^2 \mid x^2 + xy + y^2 = 1\right\}; \\ S_2 &= \left\{(x,y) \in \mathbb{R}^2 \mid e^x \sin y = e^y \sin x\right\}; \\ S_3 &= \left\{(x,y) \in \mathbb{R}^2 \mid x^2 + xy + y^2 \leqslant 1\right\}; \\ S_4 &= \left\{(x,y) \in \mathbb{R}^2 \mid x^2 + xy + y^2 < 1\right\}. \end{aligned}$$

8. Let S, T be closed subsets of $\mathbb{R}$.
(i) Show that $S \cup T$ is closed.
(ii) Show that $S \cap T$ is closed.
(iii) Give an example to show $S + T$ need not be closed.

9. Let

$$\mathcal{U} = \{(a_i, b_i) \mid i \in I\}$$

be an arbitrary collection of open intervals that 'cover' $[0,1]$; that is

$$[0,1] \subseteq \bigcup_{i \in I} (a_i, b_i).$$

This question shows that $[0,1]$ is in fact covered by finitely many of these open intervals. We define

$$S = \{x \in [0,1] : \ [0,x] \ \text{ is covered by finitely many } \ (a_i, b_i)\}.$$

(i) Show that $S \neq \varnothing$ and set $c = \sup S$.
(ii) Show that if $x \in S$ and $x < 1$, then $x + \varepsilon \in S$ for some $\varepsilon > 0$.
(iii) Deduce that $[0,1]$ is covered by finitely many (a_i, b_i).
(iv) Find a covering of $(0,1)$ by infinite open intervals, where no finite selection makes a cover.

10. (i) Let $f\colon \mathbb{R} \to \mathbb{R}$ and let S, T be closed sets such that f is continuous on both S and T. Show that f is continuous on $S \cup T$.
(ii) Show that (i) need not hold if S or T is not closed.

11. (i) Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous and $S \subseteq \mathbb{R}$. Show that $f\left(\overline{S}\right) \subseteq \overline{f(S)}$.
(ii) Give examples of continuous f and $S \subseteq \mathbb{R}$ where $f\left(\overline{S}\right) \neq \overline{f(S)}$.

12. Let d_X and d_Y be metrics on two sets X and Y respectively (Analysis I, Sheet 3, Exercise 7). We say $f\colon X \to Y$ is *continuous* if

$$\forall p \in X \quad \forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} \quad d_X(x,p) < \delta \quad \Longrightarrow \quad d_Y(f(x), f(p)) < \varepsilon.$$

(i) Let $a \in X$. Show that $f(x) = d_X(x,a)$ defines a continuous function on $f\colon X \to \mathbb{R}$.
(ii) Show that

$$D\left((x_1,y_1),(x_2,y_2)\right) = d_X(x_1,x_2) + d_Y(y_1,y_2)$$

is a metric on $X \times Y$.
(iii) Show that $f(x_1,x_2) = d_X(x_1,x_2)$ is a continuous function $f\colon X^2 \to \mathbb{R}$ using the metric D from (ii).